Exact Integer Projections for Diophantine Consistency

Research report prepared for Vladimir Reshetnikov

Explicit equations, proofs, and independently checked finite artifacts

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Abstract

We give a completely specified ordinary polynomial in 92 unrestricted integer variables, of exact total degree 692, whose absence of integer zeros is provably equivalent in ZFC to syntactic consistency of ZFC. The new construction is obtained by exact integral projections of an authenticated direct encoding of the C222 numerical proof calculus. It uses neither a K215 simulation nor an instance of Jones's universal polynomial. Every eliminated coordinate has a unique polynomial restoration with integer coefficients.

The inspected direct predecessor has 94 variables and degree 65,540. The new equation improves both of those measures, while using 29 residual squares instead of 28. Its expanded coefficient height is proved strictly below 2^{1655}; no full expansion of the principal polynomial is needed or claimed. The complete finite polynomial definition is printed in this report and supplied as a binary arithmetic graph. Lower-degree companions include a sextic in 191 integers and a sextic in 193 integers with coefficient height 16,866.

We also close a previously stated transport question for a ten-positive- variable Pell construction, and give an explicit signed, multiple-base certificate theorem. The report distinguishes new endpoints, inherited methods, a late-arriving exact duplicate, and research changes that do not improve the measured frontier. The new finite algebra is independently checked; the inherited unbounded arithmetic and logical calibration is identified rather than represented as a newly compiled proof-assistant development. Global minimality is not asserted.

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1 The result and the meaning of simplicity

1.1 The principal equation

All unqualified variables in the principal construction range over $\mathbb{Z}$. Let $S_0, \dots, S_{127} \in \mathbb{Z}[v_0, \dots, v_{190}]$ be the 128 literal residuals printed in Appendix B. Let F be the ordered set of 92 free coordinate indices, let J be the set of 29 retained residual indices, and let the 99 acyclic unit definitions be those printed in Appendix A. These finite data define polynomials $V_0(\boldsymbol{x}), \dots, V_{190}(\boldsymbol{x})$ in exactly the free inputs $\boldsymbol{x} = (x_0, \dots, x_{91})$. Define

$$\boxed{P_{92}(\boldsymbol{x}) = \sum_{r \in J} S_r(V_0(\boldsymbol{x}), \dots, V_{190}(\boldsymbol{x}))^2 = 0.} \tag{1}$$

The symbols V_j are polynomial abbreviations. They are not additional quantified unknowns. The two appendices give a complete coefficient specification of one fixed ordinary polynomial; a finite acyclic arithmetic graph provides the same specification in executable form.

Theorem 1.1 (Principal endpoint)**.** *The polynomial in* (1) *has integer coefficients and exact total degree* 692*. Its expanded coefficient height satisfies* $\mathrm{ht}(P_{92}) < 2^{1655}$*. Composing the explicit integer projections below with the retained C222/ZF/ZFC calibration gives*

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{92}\ P_{92}(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}.$$

Here Con(ZFC) *means that the specified effective ZFC presentation has no finite syntactic refutation.*

The theorem is an equivalence. It does not settle which side holds. The coefficients, coordinate lists, and arithmetic graph are chosen by finite construction and do not depend on a consistency assumption. The formalization boundary is explained in Section 4.

1.2 The comparison that is actually improved

Package 152's direct-calculus family contains a 94-integer polynomial of exact degree 65,540, presented by 28 residual squares, with the reported expanded-height bound 2^{164870} [1]. The comparison is therefore:

Measure	Inspected 94-variable form	New principal form
Independent integer variables	94	92
Exact total degree	65,540	692
Residual squares	28	29
Certified expanded-height bound	$< 2^{164870}$	$< 2^{1655}$
Largest literal scalar in definition	256	46,656

The degree is about 94.71 times smaller. The height entries are bounds, not two measured exact heights; a comparison of upper bounds is not a proof that one actual expanded height is smaller. The larger literal scalar and the extra residual are explicit costs of this particular choice of source and projection.

This is an improvement within the directly compiled proof-calculus family. The repository also contains ten-integer constructions through Sun's reduction, a 37-integer quartic, and nine-positive-variable forms with different coefficient, degree, or ancestry costs [9]. Consequently 92 is not a global minimum for the task. No argument here establishes a globally optimal polynomial under any total ordering of variables, degree, coefficient height, or description length.

1.3 Other completed endpoints

The lower-degree alternatives are fixed ordinary polynomials in unrestricted integers. Every residual coefficient is supplied, and every collected coefficient is supplied for the first six entries below.

Variables	Degree	Residuals	Collected monomials	Exact height
191	6	128	13,074	3,268,176,223
193	6	130	25,617	16,866
197	6	134	13,552	640,307
145	8	82	23,048	604,716,407,814
134	12	71	40,248	604,717,858,816
119	16	56	90,760	253,229,015,978,920,082,669,568
115	24	52	uncollected	unmeasured

For the 115-variable degree-24 form, the full residual lists are supplied but their sum of squares was not collected. Its exact degree follows from the maximum residual degree and the sum-of-squares argument below.

The 193-variable sextic is a useful alternative when small literal coefficients matter: all of its cubic residual coefficients have absolute value at most 64, and the fully collected polynomial has height $16,866$. The 191-variable sextic uses fewer variables and fewer collected monomials, with larger coefficients. These two points are different tradeoffs. The 92-variable graph should be used when the lower dimension and degree-692 profile are the desired objective.

2 Exact elimination over the integers

2.1 A unit coefficient supplies a polynomial inverse

Lemma 2.1 (Integral monic projection)**.** *Let $\epsilon \in \{-1, 1\}$, let $f \in \mathbb{Z}[\boldsymbol{x}]$, and let $R_i \in \mathbb{Z}[\boldsymbol{x}, y]$. Put*

$$A(\boldsymbol{x}, y) = (\epsilon y + f(\boldsymbol{x}))^2 + \sum_i R_i(\boldsymbol{x}, y)^2, \qquad B(\boldsymbol{x}) = \sum_i R_i(\boldsymbol{x}, -\epsilon f(\boldsymbol{x}))^2.$$

Then the integer zero sets of A and B are in bijection. The map from the smaller zero set is the integral polynomial map $\boldsymbol{x} \mapsto (\boldsymbol{x}, -\epsilon f(\boldsymbol{x}))$, and its inverse is coordinate projection. In fact

$$A(\boldsymbol{x}, -\epsilon f(\boldsymbol{x})) = B(\boldsymbol{x})$$

as a polynomial identity.

Proof. At an integer zero of A, every square is nonnegative, so each residual is zero. The distinguished row uniquely gives $y = -\epsilon f$. Conversely this value makes that row zero, and substitution gives the displayed identity. All restored coordinates are integers because ϵ is an integer unit. The two maps are inverse on the stated zero sets. □

There is no division by an arbitrary coefficient in this lemma. A row $2y - f = 0$, for example, cannot be handled by silently inserting $f/2$. Nor does the lemma authorize choosing a square root, restricting the sign of an unrestricted coordinate, or assuming that a modulus is nonzero. The checker rejects such changes by requiring the literal pivot unit.

A small example explains why degree and dimension move in opposite directions. Eliminating y from

$$(y - ab)^2 + (z - yc)^2 = 0$$

gives the sextic $(z - abc)^2 = 0$. One quantified coordinate disappears, and its multiplication becomes part of the remaining polynomial. The same operation on an actual compiled proof certificate can remove many auxiliary coordinates while preserving the complete set of integer witnesses through explicit maps.

2.2 Composition and the importance of acyclicity

Suppose several selected source rows have the form $\epsilon_v v + f_v = 0$, where f_v does not explicitly contain v. It is not enough to inspect these rows separately: the definitions could contain a cycle. For example, using both $u = v + 1$ and $v = u - 1$ as though they eliminate both coordinates would leave an unspecified free parameter.

For the principal construction the complete graph of dependencies is acyclic. Give the free coordinates their specified values, evaluate the 99 definitions in topological order, and obtain a unique integer tuple in the 191-coordinate source. The retained residuals then have their exact source values. Conversely every source zero satisfies all of these definitions. Thus the composite map is a polynomial bijection of integer zero sets, with coordinate projection as inverse.

This graph proof is independent of the heuristic that found the list of pivots. The release contains two separate checks: sequential replay of the unit eliminations and a check of the final simultaneous definitions using topological sorting. Both are exact finite algebra checks.

2.3 Degree-bounded searches and factored definitions

The lower-degree profiles were found by considering unit pivots whose actual substituted residuals remain below a specified degree bound. The search compares coordinate count, support, and coefficient costs. It uses exact integer polynomial arithmetic; randomized choices only decide which eligible pivot to try first. The final certificate fixes every choice, so verification and reconstruction are deterministic.

For larger degree bounds, fully expanding intermediate polynomials is unnecessary and can be expensive. The 92-variable definition retains the finite arithmetic graph. Degree upper bounds propagate through addition and multiplication, and a separate nonvanishing certificate proves the exact final degree. There are no variable exponents or unbounded recursion in this graph. Every node is an integer constant, a free input, an addition, a subtraction, or a multiplication.

These are applications and improvements of the repository's existing monic-elimination method [1, 3]; the general method is not claimed as a new discovery. The new work consists of the selected source combinations, optimized literal endpoints, complete restoration certificates, and their independent checks.

An initially smaller source need not lead to the best result after a degree bound is imposed. In the inspected searches, the 220-variable source exposed a better sextic projection than its more contracted 218-variable sibling. Earlier eliminations can obscure inexpensive unit pivots or increase the degree of later substitutions. The search results establish constructions, not lower bounds on what other coordinate systems could achieve.

3 The exact degree and coefficient bound

3.1 Why the principal degree is exactly 692

Propagation through the verified arithmetic graph gives maximum residual degree at most 346. Therefore $\deg P_{92} \le 692$. The release supplies an explicit integer ray $x_j = a_j t$, with all 92 values a_j recorded. On that ray, source residual 61 has coefficient

$$[t^{346}]\, S_{61}(V(\boldsymbol{a}t)) \equiv 613639 \pmod{1000003}.$$

The residue is nonzero, so the integer coefficient is nonzero. That residual has exact degree 346.

One must not now assume that the highest squares cannot cancel modulo the chosen modulus. The conclusion is instead made over the integers: if L_r is the leading degree-346 homogeneous part of a residual, then the degree-692 homogeneous part of P_{92} is $\sum_r L_r^2$. At least one L_r is nonzero, so this sum is a nonzero real polynomial. For an entirely coefficient-based proof, take the largest monomial in a multiplicative monomial order among the L_r. Its squared monomial has a positive sum of squared integer leading coefficients. Thus $\deg P_{92} = 692$.

The same argument proves exact degree $2d$ for each expanded sum-of-squares profile having a nonzero residual of maximum degree d. Nonzero modular derivative witnesses separately authenticate the use of every free variable in the principal polynomial.

3.2 Three different coefficient measures

For a polynomial $A = \sum_\alpha a_\alpha \boldsymbol{x}^\alpha$, put $\|A\|_1 = \sum_\alpha |a_\alpha|$. Then

$$\|A + B\|_1 \le \|A\|_1 + \|B\|_1 , \qquad \|AB\|_1 \le \|A\|_1 \|B\|_1 , \qquad \mathrm{ht}(A) \le \|A\|_1 .$$

Apply these inequalities to every acyclic definition and then to the sum of residual squares. The resulting explicit positive integer upper bound has 1655 bits. Hence

$$\mathrm{ht}(P_{92}) < 2^{1655}.$$

This is a rigorous upper bound on the expanded height. The number $46,656$ is instead the largest absolute literal scalar in the delivered finite definition. Neither number is a measured exact expanded height of P_{92}, and the full expanded monomial count of P_{92} is not asserted.

For the expanded companion equations the distinction is simpler: their complete collected coefficient lists are supplied, so their reported term counts and heights are exact. This is why the report labels the principal height as a bound and the companion heights as measured values.

4 Logical calibration and proof scope

The direct route starts with the retained polynomial in 225 unrestricted integers in package 158 [2]. Its source arithmetic encodes the numerical C222 proof calculus, including signed Pell certificates, finite proof-label conditions, and the associated bounded arithmetic interfaces. It does not introduce a universal-machine program number as an unspecified coefficient.

Package 167 supplies exact integer projections to the 220-variable source used here [6]. The new sparse certificate eliminates 29 coordinates to obtain the printed 191-variable sextic. The new factored certificate then eliminates 99 more coordinates to give P_{92}. The two new stages are unconditional integral polynomial maps. The separately included 193-variable sextic comes from the late 217-variable direct quartic, with its own retained parent archive.

The named logical interfaces in the inherited development include

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The supplied calibration audit checks the relation to the effective ZFC presentation and the internalizability of the argument in ZFC. That argument uses semantic model constructions, completeness, well-founded-part constructions, and the constructible inner model for the Choice bridge. This report does not replace that

argument by a claim that the entire calibration is merely finite arithmetic or has been proved in PA. No external transitivity or standardness assumption is added to the statement of Theorem 1.1.

For the new projection layer, only the elementary unit-elimination argument is needed. If the inherited source polynomial is called Q, the explicitly checked restoration gives

$$(\exists \boldsymbol{x} \in \mathbb{Z}^{92}\ P_{92} = 0) \iff (\exists \boldsymbol{y}\ Q = 0) \iff \text{there is a finite syntactic ZFC refutation.}$$

Negating the equivalence yields the stated consistency assertion.

The new package supplies ordinary mathematical proofs formalizable in ZFC and exact executable checks of the finite polynomials and maps. It does not supply a newly compiled end-to-end Lean or Rocq proof, nor a serialized object-language derivation inside ZFC. The original archives and named source interfaces are retained so that these dependencies can be audited. The finite checkers do not search for a zero, prove consistency, or replace an unbounded arithmetic theorem by testing small examples.

5 A squared transport coordinate in the compact family

The direct-calculus constructions and the following compact companion optimize different quantities. This section gives two literal polynomials in ten *strictly positive* integer variables, each of exact degree 232 and 85 binary arithmetic operations. The trace parent has 87 operations at degree 232; the inverse-Lucas parent has 85 operations at degree 304. The new squared-coordinate variant also reduces the degree of one carrier numerator from 57 to 21. The total degree remains 232.

Late comparison and attribution

At the later read-only snapshot `0d75e561a8cefc8c30fc4224605a8984c810ad1c`, the incoming Pooled Powers Separated Radicals archive already contains exactly the same 85-gate bV polynomial: all rows, all ten inputs and all four fixed coefficient values agree. The 85/232 metric point is therefore not claimed as new here. That equation is a reproduced baseline. The distinct contribution is the single replacement $L = bV \mapsto b^2$ and its soundness/completeness proof, which lowers the carrier degree without changing total degree or operation count. The exact comparison is recorded in the companion's `LATE_COMPARISON.json`.

The two parents are packages `163-trace-radical` and `164-unsquared-factors-linear-selectors` of the pinned ProveIt snapshot. The former explicitly left open the squared-coordinate transport argument. Its difficulty was a correction growing with a negative rank. Evaluating the inverse correction at a positive Pell coordinate removes that difficulty. This proves the particular new variant below; it does not assert worldwide novelty or global minimality. This compact component retains the native/K215 ancestry of its fixed coefficient tuple.

5.1 Complete polynomial

The quantified inputs are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The four fixed integers β, K_0, M_C, M_F are the complete indexed BFS943/radix67 tuple, reconstructed from the supplied finite recipe. In particular $\beta = 2^{590069} - 1$; the largest port has 592884 bits. All four actual ports, rather than arbitrary integers with the same elementary bounds, are needed for the logical

calibration. Define polynomial abbreviations

$$\begin{aligned}
h &= \beta J, & q &= h+1, & C &= q-F-\alpha,\\
X &= qw, & Y &= qs, & k &= \eta+\zeta,\\
c &= kY+\eta, & A &= Y(X+1)+1, & \Delta &= A^2-1,\\
H &= 4A-5, & v &= XY^2, & \ell &= vk,\\
E &= 1+\ell(\ell+k),
\end{aligned}$$

and

$$\begin{aligned}
R &= [h(q-F)+\alpha](q^2-1)+(M_C+qM_F)J,\\
B &= (K_0+w)C-F,\\
U &= ic^2, \qquad a=\Delta U^2E, \qquad F_0=1+a, \qquad G=EF_0,\\
f &= 1+2a, \qquad Q=4\Delta aF_0,\\
b &= cT-R, \qquad V=fb-c, \qquad K=X(3c-2X)+2,\\
e_6 &= 6(k-R-1)-\ell(k^2-4), \qquad D_6=6q^6,\\
M &= hD_6H,\\
Z &= hD_6[H(n-LB)+K]-H[D_6B+h(e_6-6q^4s)].
\end{aligned}$$

The principal weight is $L=b^2$; a companion uses $L=bV$. The equation is

$$\boxed{Q(Z^2-M^2GV^2)+(M^2G-Z^2)=0.} \tag{2}$$

Every symbol on the right is a polynomial abbreviation. There are no additional quantified variables, variable exponents, recurrence operators, or uncomputed coefficients. The literal circuit computes $\ell=(XY)(kY)$ and $f=a+F_0$, thereby sharing existing operations.

Composed with the named inherited native/refutation calibration, the ordinary mathematical theorem is

$$\mathrm{ZFC}\vdash\Big(\neg\exists\boldsymbol{x}\in\mathbb{Z}_{>0}^{10}\ P(\boldsymbol{x})=0\Big)\longleftrightarrow\mathrm{Con}(\mathrm{ZFC}).$$

The domain is essential. If nonnegative variables are desired, substitute $x_j=u_j+1$ for each of the ten inputs. This leaves the dimension and total degree unchanged, though the displayed operation count refers to the positive-input circuit.

5.2 The new transport proof

For the proof only, put

$$e_*=\frac{e_6}{6},\quad \rho=\frac{e_*}{q^6},\quad \sigma=\frac{s}{q^2},\quad t=\frac{B}{h},\quad \kappa=\frac{K}{H},\quad W=1+hL,\quad z=\frac{Z}{M}.$$

No integrality of these quotients is initially assumed. The exact rational identity is

$$z=n+\kappa+\sigma-Wt-\rho. \tag{3}$$

The trace parent's root extraction and low-index exclusions depend on Q,V,G and the final norm, not on L or e_6. At a positive zero they therefore still give

$$G=g^2,\quad z=\varepsilon gy,\quad QV^2-(Q-1)y^2=1,\qquad g,y>0,\quad \varepsilon\in\{-1,1\},$$

with z, y integral, and

$$b \neq 0, \quad |b| > 8f^3 - \tfrac{1}{2}, \quad g < f/2, \quad y < 3|V|/2, \quad c < f/2.$$

These statements are established before any sign assumption on R or B. In particular, since b is a nonzero integer,

$$|V| \leq f|b| + c < \tfrac{3}{2}f|b|, \qquad gy < \tfrac{9}{8}f^2|b| < \tfrac{1}{2}b^2.$$

The last strict inequality follows from $|b| > 8f^3 - 1/2 > (9/4)f^2$ for $f \geq 1$. Consequently $L = b^2$ satisfies

$$L > 0, \qquad gy < L/2, \qquad |b|/q^6 < L/2.$$

The inverse correction has the decisive sign:

$$\frac{vk(k^2-4)}{6} \geq 0, \qquad e_* \leq k - R - 1 < b,$$

because $k \geq 2$ and $b - (k - R - 1) = cT - k + 1 > 0$. This holds for every integer R, including negative ranks. Thus $\rho < |b|/q^6$. If $B \leq -1$, then $Wt < -L$; using the positive n, κ, σ in (3) gives

$$n = \varepsilon gy - \kappa - \sigma + Wt + \rho < L/2 - L + |b|/q^6 < 0,$$

a contradiction. Therefore $B \geq 0$ and then $C > 0$. This is the new unbounded sign argument. It is not inferred from bounded testing or from a formal substitution into the forward-correction proof.

5.3 The remaining arithmetic order

Once $C > 0$, the inherited mask bound gives $3q < R < q^4 - 2$. The unsquared trace obeys $f \equiv 1 \pmod{c}$, so the two auxiliary residues remain $V \equiv -c \pmod{f}$ and $V \equiv -R \pmod{c}$. The retained signed-index theorem authenticates the main Pell rank as R.

For clarity, the inverse denominator argument must retain its order. The first Pell recurrence has

$$k = 2m + 8vC_m + 32v^2Z_m, \qquad C_m = \binom{m+1}{3}, \quad Z_m \in \mathbb{N}.$$

Putting $d = 2m - R - 1$, $A_m = C_m + 4vZ_m$ and $S_m = k^2 + 2mk + 4m^2 - 4$ gives the exact identity

$$3(e_* - d) = 4v^2(24Z_m - A_mS_m).$$

The carrier gives $q^6 \mid (e_* - q^4 s)$, hence $q^4 \mid e_*$. Since $v = q^3ws^2$, prime valuations in the displayed identity give $q^4 \mid (e_* - d)$ even at the prime 3. The bounds $|d| < R < q^4$ then force $d = 0$.

It would be premature to cancel the remaining factor 3 and infer $q^6 \mid e_*$. Instead, the carrier first gives $H \mid 6hK$. The retained uniform ratio estimate and whole-product projection force $X = 2^R$. Since $q \mid X$, the radix q is a power of two. Only now does the identity imply $q^6 \mid e_*$. The carrier then gives $q^2 \mid s$ and $h \mid B$, supplying the exact native outer endpoint. The parent's indexed native/refutation calibration completes soundness.

Conversely, a finite refutation gives the canonical outer witnesses and the trace parent's positive auxiliary Pell pair. Its actual positive integer input is $T = (V + c + Rf)/(cf)$. The inverse error can be negative, but the inherited explicit bounds give

$$|\rho| < vk^3/3 < g/3, \qquad \kappa + \sigma < g/2.$$

For either positive weight choose

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma.$$

This is an integer and is strictly positive because $y \geq 1$, $t \geq 0$ and $g > \kappa + \sigma + |\rho|$. The carrier now has $z = gy$, proving completeness for all ten positive inputs.

5.4 Exact complexity and scope

Let $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $B_0 = k_0 sT - q_0^2(q_0 - F)$. The complete highest homogeneous component is

$$[P]_{232} = -9\,2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_0^2.$$

It is nonzero at the actual $\beta > 0$. For $L = b^2$, the degrees of (Q, M, G, V, Z) are $(80, 11, 50, 40, 21)$. The term QZ^2 has degree 122, while $-QM^2GV^2$ has degree 232 and remains the unique dominant summand. For $L = bV$, $\deg Z = 57$ and $\deg(QZ^2) = 194$.

The reproducible finite checker covers every live gate by 36 exact free-polynomial cuts per graph, checks the complete 202-monomial highest component and all four coefficient payloads, and rejects 22 deliberately corrupted operators per graph. Normal and optimized Python runs agree. The finite recipe reconstructs every fixed integer and matches its pinned raw SHA-256 identity. These are algebraic artifact checks, separate from the unbounded arguments and inherited logical calibration. No new end-to-end proof-assistant development is claimed.

5.5 A smaller, separately calibrated coefficient tuple

Both 85-operation, degree-232 forms also admit the complete inherited *complemented-label radix47* specialization. All four fixed ports change together, while every free-port gate remains identical. This separate specialization has

$$\beta = 2^{413929} - 1, \qquad \text{largest port} = 415904 \text{ bits}, \qquad \text{total raw bytes} = 207214.$$

The radix67 tuple has 295388 raw bytes, so the inherited smaller tuple saves 88174 bytes in this matched storage measure. Its coefficient reduction is credited to its native predecessor; the present contribution is its explicit composition with the new 85/232 polynomials.

This is not a coefficient-storage record. The late incoming comparison already supplies a separately calibrated one-anchor7309 tuple using 172011 raw bytes. Its bV graph also duplicates the 85/232 baseline, as disclosed above. The present distinct carrier change remains $L = b^2$.

This is the whole complemented-label tuple with seven shared marker weights, not the conditional47 layout and not the radix67 formulas with a substituted radix. The complete ordinary carry/controller proof, its native bridge, the 3297-coefficient array, all 51 native target identities and the literal native P66 polynomial are retained and authenticated against the bridge's original source hashes. The finite recipe reconstructs every port and checks the actual controller at prime 1012691, its exact orders and its valuation-one lift.

The new trace and inverse-correction arguments are uniform in the four ports subject to the matched scalar bounds, mask phases and population identities. They recover the same outer endpoint with $W_{\text{base}} = w$. The whole47 bridge then reconstructs fresh source witnesses at $A_{\text{source}} = Y(X + 1) + 2$ and applies its own complemented-label decoder. Conversely its own controller supplies the canonical positive outer data; the new bound $g > \kappa + \sigma + |\rho|$ proves the constructed n positive. Thus both directions use this actual whole tuple. No old67 decoder or pointwise identification of the two specialized polynomials is assumed.

The complete alternate coefficient recipe is supplied separately with the radix47 component. Its independent verifier checks the whole binding, all live gates, full degree-232 leading forms, and actual corruptions. It carries the same explicitly stated ordinary-proof and formalization boundary as the main compact companion.

6 Oriented rational error certificates

The incoming reciprocal-pooling construction combines two consecutive bases [7]. Its final research paragraph already suggests oriented weights for several signed bases. The following result closes the size

and zero-base details for five blocks and is implemented in the supporting 209-variable degree-24 equation. That endpoint is not competitive with all the exact-projection profiles above; its value here is the completed arithmetic certificate and its explicit compiler.

Theorem 6.1 (Five signed blocks)**.** *Suppose $W \geq 258$, $|A| > W^W$, and there are at most five integer bases n_j, each satisfying $n_j^2 < W$, and at most 17 requests altogether. For each request let $m_i, k_i \in \mathbb{Z}$ satisfy $m_i^2, k_i^2 < W$, and let (X_i, Y_i) be the canonical signed Pell coordinates at parameter A and index k_i. Set*

$$T_j = 2An_j - n_j^2 - 1, \quad D_i = m_i - X_i + AY_i,$$

$$H_j = \sum_{i \in I_j} Y_i(m_i - X_i), \qquad E_j = \sum_{i \in I_j} (D_i^2 - Y_i^2) - 2n_j H_j.$$

Then the existence of an integer z satisfying

$$\sum_j n_j E_j \prod_{\ell \neq j} T_\ell - z \prod_j T_j = 0 \tag{4}$$

implies $m_i = n_j^{k_i}$ for every block with $n_j \neq 0$. At nonunit bases it also implies $k_i \geq 0$. Conversely, these true powers in all nonzero-base blocks imply the existence of such a z. Zero-base blocks impose no condition in this theorem.

Proof. For a nonzero base, $|T_j| > W^W$ and $\operatorname{sign}(T_j) = \operatorname{sign}(An_j)$. For a zero base, $T_j = -1$, so none of the denominators is zero. Put $\delta_i = D_i - n_j Y_i$. Direct expansion gives

$$\sum_{i \in I_j} \delta_i^2 = E_j - T_j \sum_{i \in I_j} Y_i^2.$$

Consequently (4) makes $\sum_j n_j \sum_i \delta_i^2 / T_j$ an integer.

Let $b = \lfloor \sqrt{W - 1} \rfloor \geq 16$ and take the product only over nonzero bases in $C = \prod_{n_j \neq 0} n_j^b$. For each request in such a block define

$$d_i = Cm_i - (C/n_j^b) n_j^{b+k_i} \in \mathbb{Z}.$$

The exponents are nonnegative and the indicated quotient is an integer. The canonical Pell congruence gives $C\delta_i \equiv d_i \pmod{T_j}$. It remains valid for negative indices because $\gcd(n_j, T_j) = 1$. It follows that $\sum_j n_j \sum_i d_i^2 / T_j$ is an integer.

There are at most five nonzero factors in C. Thus $|d_i| \leq 2b^{6b}$ and

$$\sum_j |n_j| \sum_{i \in I_j} d_i^2 \leq 68 b^{12b+1} < W^{6b+2} \leq W^W.$$

Here $b^2 < W$, $68b < W^2$, and $6b + 2 \leq W$ for $b \geq 16$. The integer rational sum has absolute value below one. Moreover every nonzero weight n_j / T_j has the same sign, namely the sign of A. Hence that integer is zero and every d_i is zero. Dividing in $\mathbb{Q}$ by the nonzero integer C proves the power identities. Negative indices at a nonunit base cannot give an integer output.

Conversely true powers give $T_j \mid \delta_i$, whence $T_j \mid E_j$. The value $z = \sum_{n_j \neq 0} n_j E_j / T_j$ is an integer and satisfies (4). The terms from zero bases are zero. □

The actual five bases are $F^2 + 1$, B, BH, q, and $q + 1$. Their nonvanishing is recovered in a checked order: the retained norm forces $M \neq 0$; the relation $Bv = M$ forces $B \neq 0$; oddness gives $H \neq 0$. The first three blocks can now be interpreted. The retained rounding bound gives $|B| \geq 2$, and the exact powers give $q = (BH)^x \neq 0, -1$. Therefore the two remaining blocks are also covered. This avoids assuming the later radix positivity in order to authenticate the earlier power conditions.

The supporting compiler removes ten private certificate coordinates and introduces one integer z in the protected 218-variable source. The resulting 209-variable polynomial has 145 quadratic residuals and one residual of degree 12, hence exact total degree 24. The independent checker authenticates the protected row basis and every replacement. No expanded support or coefficient-height record is claimed for it.

7 A sound semantic alternative that does not improve the frontier

The supplementary factored-radix construction replaces the request $q = (BH)^x$ by

$$N = B^x, \qquad P = H^x, \qquad q = NP.$$

Its changed frame uses 35 semantic atom squares. The outputs and indices retain squared bounds by W, while the two bases $q, q+1$ need only the weaker bounds $|q|, |q+1| < W$. The same-base certificate remains sound: with $b = \lfloor \sqrt{W-1} \rfloor$, the cleared error bound becomes

$$60W^{4b} < W^{4b+1} \le W^W, \qquad |2An - n^2 - 1| \ge 2|A| - W^2 > W^W.$$

The full proof checks the signed cases and the ordering of witness choices. In particular the new frame contains only semantic coordinates chosen before the large Pell parameter, not private Pell witnesses whose size depends on that parameter.

The literal source has 224 variables, 161 residuals, and degree six; five rechecked radical projections give a 219-variable source. The subsequent searches reached 193-variable sextics, so this changed semantics has not improved the frozen 191-variable sextic frontier. It is included as completed research with that limitation, not counted as an unachieved 190-variable result.

8 Verification, provenance, and integration

8.1 What was checked

The principal component binds its 191-variable source to the exact 220-to-191 endpoint, checks all 99 unit definitions and their acyclic dependencies, checks all retained residuals, and supplies a literal binary arithmetic graph. It independently verifies the degree witness, the height bound, and actual occurrence of the free coordinates. The integer restoration can be evaluated at arbitrary supplied tuples. The resulting value of the source sum of squares agrees exactly with the value of the projected polynomial.

For the sparse companions, a separately implemented verifier uses exponent maps rather than the producer's repeated-index representation. It substitutes through every remaining row, checks the full endpoint, renumbers coordinates, and compares every collected coefficient where an expansion is supplied. Deliberate pivot, restoration, and endpoint corruptions must be rejected, including under optimized Python. The compact and rational-pooling components have their own exact identity and source-binding checks. Repeated small numerical tests are not offered as proof of an unbounded assertion.

The arithmetic-parent replays and the exact scope of each fresh run are recorded in the verification receipts. Included historical receipts are not represented as fresh runs of their entire transitive proof development. Independent reviews in this report are separate tasks within an AI-assisted research workflow, not outside human refereeing.

8.2 Immutable comparison boundaries

The initial inspection was pinned to commit `35fa5f6bc659e83f2b2034126c4bcf8e023fbe1c`. The repository changed during the work. A late inspection at `0d75e561a8cefc8c30fc4224605a8984c810ad1c`

covered newly placed packages 166–168 and three new incoming archives. It found direct quartics with 216 or 217 variables and an exact duplicate of the compact bV 85-operation degree-232 graph. Those findings are included in the comparisons and source ledger.

In particular the compact 85/232 metric is not claimed as a new record, and the smaller historical radix47 coefficient tuple is not described as the current storage record. The distinct compact result here is the proved b^2 carrier. The new direct projection endpoints are compared with the identified sources, rather than an unspecified moving repository head.

8.3 Using the archive

The principal equation, its binary graph, source, certificates, evaluator, and independent checker are under `principal92/`. The lower-degree ordinary equations are under `equations/`. All these definitions use literal integers and finite operations. The original archives retained under `inherited/` preserve the arithmetic and logical proof dependencies and their notices.

From the extracted release directory, the principal commands are:

```
python3 -B principal92/verify.py
python3 -B verify.py
python3 -B -O verify.py
python3 -B verify.py --components
```

Each successful command checks a finite artifact. It is not a procedure for deciding whether the polynomial has an integer zero. The README gives exact evaluation and restoration commands, the report build instructions, and optional parent-replay commands.

For additive integration into ProveIt, keep the literal equation and the source-binding certificates together. A useful formalization target is the generic integral monic-projection theorem plus a checked interpreter for the finite graph. The inherited C222 and ZF/ZFC calibration remains a separately identifiable dependency.

A Complete finite definition of the 92-variable polynomial

The next data, together with the source rows in Appendix B, specify (1) without a coefficient oracle or an unspecified program parameter. Let the increasing free list be $F = (F_0, \dots, F_{91})$. Set $V_{F_j} = x_j$. For each listed triple (r, v, ϵ) define the polynomial

$$f_{r,v}(\boldsymbol{u}) = -\epsilon(S_r(\boldsymbol{u}) - \epsilon u_v), \qquad V_v = f_{r,v}(V_0, \dots, V_{190}).$$

The displayed source row contains u_v only as the term ϵu_v. The table is ordered so that dependencies on other defined coordinates precede their use. It therefore specifies polynomial abbreviations, not simultaneous existential equations. Finally square and sum the rows indexed by J.

Free coordinates and retained residuals

$$\begin{aligned} F = (&4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 17, 18, 20, 42,\\ &46, 50, 78, 80, 81, 82, 84, 85, 87, 88, 90, 91, 93, 94, 96, 97,\\ &98, 100, 101, 103, 105, 107, 108, 110, 111, 113, 114, 117, 121, 123, 124, 125,\\ &126, 127, 129, 130, 132, 133, 135, 136, 138, 139, 141, 142, 144, 145, 147, 148,\\ &150, 151, 154, 156, 157, 159, 160, 162, 163, 166, 167, 171, 172, 173, 174, 175, \end{aligned}$$

$$
\begin{aligned}
&178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 189, 190)\\
J = (&13, 27, 31, 35, 61, 77, 79, 84, 86, 88, 90, 92, 94, 96, 98, 100,\\
&102, 104, 106, 108, 112, 114, 119, 121, 122, 123, 124, 126, 127)
\end{aligned}
$$

The 99 unit definitions in dependency order

r	v	ϵ	r	v	ϵ	r	v	ϵ	r	v	ϵ
110	0	1	4	19	1	5	21	1	6	22	1
7	23	1	18	33	1	32	47	1	36	51	1
47	62	1	120	104	1	115	168	1	2	1	1
1	2	-1	9	25	1	11	27	1	12	28	1
14	29	1	16	31	1	22	37	1	33	48	1
38	53	1	48	63	1	65	86	-1	66	89	-1
67	92	-1	68	95	-1	69	99	-1	70	102	-1
71	106	-1	72	109	-1	73	112	-1	74	115	-1
116	116	1	0	3	1	8	24	1	15	30	1
17	32	1	19	34	1	21	36	1	23	38	1
34	49	1	37	52	1	40	55	1	51	65	1
78	122	1	83	128	1	85	131	1	87	134	1
89	137	1	91	140	1	93	143	1	95	146	1
97	149	1	99	152	1	81	153	-1	101	155	1
103	158	1	105	161	1	107	164	1	111	176	1
125	188	-1	3	16	1	10	26	1	20	35	1
24	39	1	28	43	1	39	54	1	41	56	-1
42	57	-1	43	58	-1	44	59	-1	45	60	-1
46	61	-1	50	66	1	52	67	1	80	120	-1
117	169	1	25	40	1	29	44	1	49	64	1
53	68	1	60	75	1	62	76	1	63	77	1
109	165	1	26	41	1	30	45	1	54	69	1
82	79	1	118	170	1	55	70	1	64	83	-1
75	118	1	113	177	1	56	71	1	76	119	1
57	72	1	58	73	1	59	74	1			

B All 128 literal source residuals

These are the complete residuals $S_r(\boldsymbol{v})$ of the 191-variable sextic source, indexed from zero. Every coefficient and exponent below is literal. The source equation is $\sum_{r=0}^{127} S_r(\boldsymbol{v})^2 = 0$.

$$
\begin{aligned}
S_0 &= -v_2^2 + v_3\\
S_1 &= v_0v_5v_7 - v_0v_7 - v_2 - v_5v_6 + v_5v_7\\
S_2 &= -1 - v_0v_5v_7 + v_0v_7 + v_1 - v_5v_7\\
S_3 &= 1 + v_3 - v_4v_{18} + v_{16}\\
S_4 &= -1 + 2v_8 - v_8^2 - 2v_8v_9 + 2v_9 - v_9^2 + v_{19}\\
S_5 &= -2 + 2v_8 - v_8^2 - 4v_8v_{20} + 4v_{20} - 4v_{20}^2 + v_{21}\\
S_6 &= -2 - v_9^2 - 2v_9v_{10} - v_{10}^2 + v_{22}
\end{aligned}
$$

$$
\begin{aligned}
S_{7} =& -1+2v_{8}-v_{8}^{2}-2v_{8}v_{10}+2v_{10}-v_{10}^{2}+v_{23}\\
S_{8} =& \,19-26v_{8}-29v_{8}^{2}-34v_{8}v_{9}-36v_{8}v_{10}-24v_{8}v_{19}-24v_{8}v_{23}\\
&+41v_{9}+38v_{9}^{2}-48v_{9}v_{10}+16v_{9}v_{22}-28v_{9}v_{23}+36v_{10}+70v_{10}^{2}\\
&-32v_{10}v_{19}+32v_{10}v_{22}-13v_{19}-8v_{19}v_{23}-v_{20}+40v_{22}-14v_{23}\\
&+v_{24}-v_{27}\\
S_{9} =& -6+16v_{10}-16v_{10}^{2}-16v_{10}v_{22}-16v_{10}v_{23}+8v_{22}-4v_{22}^{2}-8v_{22}v_{23}\\
&+8v_{23}-4v_{23}^{2}+v_{25}\\
S_{10} =& \,6+36v_{8}-48v_{8}v_{10}-48v_{8}v_{23}-24v_{8}v_{25}+48v_{9}-64v_{9}v_{10}\\
&-64v_{9}v_{23}-32v_{9}v_{25}+7v_{10}+9v_{10}^{2}-16v_{10}v_{19}+8v_{10}v_{23}\\
&+4v_{10}v_{25}+12v_{19}-16v_{19}v_{23}-8v_{19}v_{25}-v_{20}+8v_{23}+4v_{25}\\
&+v_{26}-v_{36}\\
S_{11} =& \,96-29v_{8}-37v_{8}^{2}-24v_{8}v_{9}-112v_{8}v_{10}-4v_{8}v_{22}-24v_{8}v_{23}\\
&-39v_{9}+17v_{9}^{2}-4v_{9}v_{10}-24v_{9}v_{22}-4v_{9}v_{23}-80v_{10}-74v_{10}^{2}\\
&-32v_{10}v_{22}-32v_{10}v_{23}-66v_{22}-4v_{22}^{2}-8v_{23}-4v_{23}^{2}+v_{27}\\
S_{12} =& -12+16v_{8}-4v_{8}^{2}-8v_{8}v_{10}-8v_{8}v_{19}-8v_{8}v_{22}+16v_{10}-4v_{10}^{2}\\
&-8v_{10}v_{19}-8v_{10}v_{22}+16v_{19}-4v_{19}^{2}-8v_{19}v_{22}+16v_{22}-4v_{22}^{2}\\
&+v_{28}\\
S_{13} =& -4-30v_{8}-8v_{8}^{2}-64v_{8}v_{9}+16v_{8}v_{19}-48v_{9}-64v_{9}^{2}-6v_{19}\\
&+8v_{19}^{2}+2v_{20}-v_{31}\\
S_{14} =& -82-414v_{8}-529v_{8}^{2}-1472v_{8}v_{9}-368v_{8}v_{19}+92v_{8}v_{20}-576v_{9}\\
&-1024v_{9}^{2}-512v_{9}v_{19}+128v_{9}v_{20}-144v_{19}-64v_{19}^{2}+32v_{19}v_{20}\\
&+36v_{20}-4v_{20}^{2}+v_{29}\\
S_{15} =& -1453-10332v_{8}-17139v_{8}^{2}-46656v_{8}v_{9}-11664v_{8}v_{19}\\
&+2916v_{8}v_{20}+744v_{8}v_{29}-14016v_{9}-31744v_{9}^{2}-15872v_{9}v_{19}\\
&+3968v_{9}v_{20}+1024v_{9}v_{29}-3504v_{19}-1984v_{19}^{2}+992v_{19}v_{20}\\
&+256v_{19}v_{29}+876v_{20}-124v_{20}^{2}-64v_{20}v_{29}+225v_{29}-4v_{29}^{2}+v_{30}\\
S_{16} =& -1-28v_{8}^{2}-32v_{8}v_{9}-40v_{8}v_{19}+8v_{9}-32v_{9}v_{19}-4v_{19}-12v_{19}^{2}\\
&-v_{20}+v_{31}\\
S_{17} =& \,49-34v_{9}^{2}-44v_{9}v_{10}-24v_{9}v_{22}-32v_{9}v_{31}+6v_{10}-10v_{10}^{2}\\
&-24v_{10}v_{22}-24v_{10}v_{31}-32v_{22}-8v_{22}v_{31}+4v_{31}-4v_{31}^{2}+v_{32}\\
&-v_{37}\\
S_{18} =& -v_{10}^{2}-2v_{10}v_{11}-v_{11}^{2}+v_{33}\\
S_{19} =& -26+14v_{8}+v_{8}^{2}-8v_{8}v_{9}+8v_{8}v_{10}-8v_{8}v_{22}-8v_{8}v_{23}\\
&-4v_{8}v_{33}+8v_{9}^{2}+16v_{9}v_{22}-8v_{9}v_{23}-8v_{10}+2v_{10}^{2}-8v_{10}v_{33}\\
&+8v_{22}^{2}-8v_{22}v_{23}+18v_{23}-4v_{23}^{2}+4v_{33}-4v_{33}^{2}+v_{34}-3v_{37}\\
S_{20} =& \,4v_{8}-18v_{8}^{2}-40v_{8}v_{10}-32v_{8}v_{23}-40v_{8}v_{33}-16v_{8}v_{34}\\
&+4v_{10}-24v_{10}^{2}-32v_{10}v_{23}-48v_{10}v_{33}-16v_{10}v_{34}-16v_{23}^{2}\\
&-32v_{23}v_{33}-16v_{23}v_{34}+4v_{33}-24v_{33}^{2}-16v_{33}v_{34}-4v_{34}^{2}+v_{35}\\
S_{21} =& \,v_{10}-25v_{10}^{2}-40v_{10}v_{23}-20v_{10}v_{25}-16v_{23}^{2}-16v_{23}v_{25}-4v_{25}^{2}\\
&+v_{36}\\
S_{22} =& -1+8v_{9}-4v_{9}^{2}-8v_{9}v_{22}+8v_{22}-4v_{22}^{2}+v_{37}\\
S_{23} =& -1-16v_{9}^{2}-8v_{9}v_{10}-32v_{9}v_{22}-16v_{9}v_{37}-v_{10}^{2}-8v_{10}v_{22}\\
&-4v_{10}v_{37}-16v_{22}^{2}-16v_{22}v_{37}-4v_{37}^{2}+v_{38}
\end{aligned}
$$

$$
\begin{aligned}
S_{24} &= 3 + v_{9} - 2v_{9}v_{10} - 4v_{9}v_{38} - 3v_{10}^{2} - 8v_{10}v_{38} - v_{22} - 4v_{38}^{2} \\
&\quad + v_{39} \\
S_{25} &= -1 - 4v_{39} - 4v_{39}^{2} + v_{40} \\
S_{26} &= -1 - v_{10}^{2} - 8v_{10}v_{39} - 4v_{10}v_{40} - 16v_{39}^{2} - 16v_{39}v_{40} - 4v_{40}^{2} + v_{41} \\
S_{27} &= 4 + v_{9} - 2v_{9}v_{10} - 4v_{9}v_{41} - 3v_{10}^{2} - 8v_{10}v_{41} - v_{22} - 4v_{41}^{2} \\
&\quad + v_{42} \\
S_{28} &= -4v_{36}^{2} - 8v_{36}v_{42} - 4v_{42}^{2} + v_{43} \\
S_{29} &= -9 + 7v_{8} - v_{8}^{2} - 8v_{8}v_{36} - 4v_{8}v_{43} + 24v_{36} - 16v_{36}^{2} - 16v_{36}v_{43} \\
&\quad + 12v_{43} - 4v_{43}^{2} + v_{44} \\
S_{30} &= -1 - 4v_{44} - 4v_{44}^{2} + v_{45} \\
S_{31} &= -v_{10}^{2} - 8v_{10}v_{44} - 4v_{10}v_{45} - 16v_{44}^{2} - 16v_{44}v_{45} - 4v_{45}^{2} + v_{46} \\
S_{32} &= -v_{8} - v_{8}^{2} - 4v_{8}v_{10} - 4v_{8}v_{46} - 4v_{10} - 4v_{10}^{2} - 8v_{10}v_{46} - 4v_{46} \\
&\quad - 4v_{46}^{2} + v_{47} \\
S_{33} &= -10 - 12v_{47} - 4v_{47}^{2} + v_{48} \\
S_{34} &= 2 - 4v_{9} - 16v_{9}v_{47} - 8v_{9}v_{48} - 4v_{22} - 16v_{22}v_{47} - 8v_{22}v_{48} \\
&\quad - v_{37} + 8v_{47} - 16v_{47}^{2} - 16v_{47}v_{48} + 4v_{48} - 4v_{48}^{2} + v_{49} \\
S_{35} &= -10 + v_{9} - 25v_{9}^{2} - 40v_{9}v_{22} - 20v_{9}v_{49} - 16v_{22}^{2} - 16v_{22}v_{49} \\
&\quad - 12v_{42} - 4v_{42}^{2} - 4v_{49}^{2} + v_{50} \\
S_{36} &= 21 + 24v_{42} - 8v_{42}^{2} - 16v_{42}v_{50} - 2v_{50} - 4v_{50}^{2} + v_{51} \\
S_{37} &= 1 + 4v_{8} - 4v_{8}^{2} - 8v_{8}v_{19} - 8v_{8}v_{53} + v_{10} - v_{10}^{2} - 4v_{10}v_{51} \\
&\quad + 4v_{19} - 4v_{19}^{2} - 8v_{19}v_{53} - 4v_{51}^{2} + v_{52} + 4v_{53} - 4v_{53}^{2} \\
S_{38} &= -4v_{8} - 4v_{8}^{2} - 4v_{8}v_{10} - 8v_{8}v_{19} - v_{10} - v_{10}^{2} - 4v_{10}v_{19} - 4v_{19} \\
&\quad - 4v_{19}^{2} + v_{53} \\
S_{39} &= 4v_{8} - 24v_{8}^{2} - 48v_{8}v_{19} - 16v_{8}v_{52} - 16v_{8}v_{53} + 4v_{19} - 24v_{19}^{2} \\
&\quad - 16v_{19}v_{52} - 16v_{19}v_{53} - 4v_{52}^{2} + 8v_{53} - 8v_{53}^{2} + v_{54} \\
S_{40} &= -v_{2}v_{13} + v_{55} \\
S_{41} &= -6v_{8}^{2}v_{55} - 120v_{8}v_{20}v_{55} - 48v_{8}v_{21}v_{55} - 120v_{8}v_{55} + v_{12} \\
&\quad - v_{13} - 248v_{20}^{2}v_{55} - 128v_{20}v_{21}v_{55} - 232v_{20}v_{55} - 8v_{21}^{2}v_{55} \\
&\quad - 62v_{21}v_{55} + 134v_{55} - v_{56} \\
S_{42} &= -4v_{8}^{2}v_{55} - 16v_{8}v_{9}v_{55} - 16v_{8}v_{22}v_{55} + 28v_{8}v_{55} + 8v_{9}^{2}v_{55} \\
&\quad + 16v_{9}v_{22}v_{55} + v_{12} - v_{13} + 8v_{22}^{2}v_{55} - 8v_{24}^{2}v_{55} - 6v_{37}v_{55} \\
&\quad - 30v_{55} - v_{57} \\
S_{43} &= -4v_{8}^{2}v_{55} - 48v_{8}v_{9}v_{55} - 64v_{8}v_{10}v_{55} - 16v_{8}v_{22}v_{55} \\
&\quad - 20v_{8}v_{55} - 36v_{9}^{2}v_{55} - 168v_{9}v_{10}v_{55} - 96v_{9}v_{22}v_{55} \\
&\quad - 144v_{9}v_{55} - 148v_{10}^{2}v_{55} - 128v_{10}v_{22}v_{55} - 192v_{10}v_{55} + v_{12} \\
&\quad - v_{13} - 16v_{22}^{2}v_{55} - 156v_{22}v_{55} - 8v_{27}^{2}v_{55} + 180v_{55} - v_{58} \\
S_{44} &= -96v_{8}^{2}v_{55} - 256v_{8}v_{10}v_{55} - 128v_{8}v_{22}v_{55} - 64v_{8}v_{23}v_{55} \\
&\quad - 64v_{8}v_{28}v_{55} - 112v_{8}v_{55} - 104v_{10}^{2}v_{55} - 96v_{10}v_{22}v_{55} \\
&\quad - 16v_{10}v_{23}v_{55} - 80v_{10}v_{28}v_{55} - 180v_{10}v_{55} + v_{12} - v_{13} \\
&\quad - 16v_{22}^{2}v_{55} - 32v_{22}v_{28}v_{55} - 48v_{22}v_{55} + 8v_{23}^{2}v_{55} - 16v_{23}v_{28}v_{55} \\
&\quad - 76v_{23}v_{55} - 4v_{25}v_{55} - 8v_{28}^{2}v_{55} - 8v_{28}v_{55} + 56v_{55} - v_{59} \\
S_{45} &= -2v_{9}^{2}v_{55} - 8v_{9}v_{30}v_{55} + v_{12} - v_{13} - 8v_{30}^{2}v_{55} - 4v_{30}v_{55}
\end{aligned}
$$

$$
\begin{aligned}
&\quad - v_{60} \\
S_{46} = &- 128v_8^2v_{55} - 192v_8v_{10}v_{55} - 64v_8v_{23}v_{55} - 64v_8v_{32}v_{55} \\
&+ 16v_8v_{55} - 72v_{10}^2v_{55} - 48v_{10}v_{23}v_{55} - 48v_{10}v_{32}v_{55} \\
&+ 12v_{10}v_{55} + v_{12} - v_{13} - 8v_{23}^2v_{55} - 16v_{23}v_{32}v_{55} + 4v_{23}v_{55} \\
&- 8v_{32}^2v_{55} + 8v_{32}v_{55} - v_{61} \\
S_{47} = &- v_8v_9 + v_8v_{10} + v_9 + v_9v_{10} - v_{10} - v_{10}^2 + v_{62} \\
S_{48} = &\, 1 + 2v_{11} + v_{11}^2 - v_{62}^2 + v_{63} \\
S_{49} = &- 8v_{54}^2v_{55} - 12v_{54}v_{55} - 2v_{55} + v_{64} \\
S_{50} = &\, 2v_2v_8v_{65} + 2v_2v_{19}v_{65} - 2v_2v_{65} - v_9v_{55} - v_{13} + v_{65} \\
&+ v_{66} \\
S_{51} = &- v_2v_8v_{12} + v_2v_{12} - v_{12} + v_{65} \\
S_{52} = &\, v_2v_9v_{12} - 6v_8v_{55} - 8v_9v_{55} + v_{12} - v_{13} - 2v_{19}v_{55} \\
&+ v_{67} \\
S_{53} = &- v_{12}v_{56}v_{57} + v_{13}v_{56}v_{57} + v_{68} \\
S_{54} = &- v_{12}v_{68} + v_{13}v_{68} + 2v_{26}v_{55}v_{68} + 6v_{55}v_{68} + v_{69} \\
S_{55} = &- v_{58}v_{59}v_{69} + v_{70} \\
S_{56} = &- v_{60}v_{61}v_{70} + v_{71} \\
S_{57} = &- v_{12}v_{71} + v_{13}v_{71} + 2v_{35}v_{55}v_{71} + 4v_{55}v_{71} + v_{72} \\
S_{58} = &- v_{66}v_{67}v_{72} + v_{73} \\
S_{59} = &- v_{73}^2 + v_{74} \\
S_{60} = &- v_5v_{14} - v_5v_{14}v_{16} + v_{14}v_{16} + v_{75} \\
S_{61} = &- 2v_1^2 + 2v_1v_{12} + 2v_1v_{13} + 2v_1v_{75} + 2v_1v_{76} - v_{12}^2 \\
&- v_{12}^2v_{74} + 2v_{12}v_{13}v_{74} + 2v_{12}v_{64}v_{74} - 2v_{12}v_{75} - v_{13}^2 \\
&- v_{13}^2v_{74} - 2v_{13}v_{64}v_{74} - 2v_{13}v_{76} + v_{17}v_{78} - v_{63}^2v_{74} \\
&- v_{64}^2v_{74} - v_{75}^2 - v_{76}^2 \\
S_{62} = &- 2v_5v_{15} - v_5v_{15}v_{16} + v_{15} + v_{15}v_{16} + v_{76} \\
S_{63} = &\, 330 - 217v_8 - 238v_8^2 - 64v_8v_9 - 512v_8v_{10} - 120v_8v_{20} \\
&- 48v_8v_{21} - 160v_8v_{22} - 128v_8v_{23} - 64v_8v_{28} - 64v_8v_{32} \\
&- 153v_9 - 30v_9^2 - 168v_9v_{10} - 80v_9v_{22} - 8v_9v_{30} - 360v_{10} \\
&- 324v_{10}^2 - 224v_{10}v_{22} - 64v_{10}v_{23} - 80v_{10}v_{28} - 48v_{10}v_{32} - 4v_{19} \\
&- 232v_{20} - 248v_{20}^2 - 128v_{20}v_{21} - 62v_{21} - 8v_{21}^2 - 204v_{22} - 24v_{22}^2 \\
&- 32v_{22}v_{28} - 72v_{23} - 16v_{23}v_{28} - 16v_{23}v_{32} - 8v_{24}^2 - 4v_{25} - 2v_{26} \\
&- 8v_{27}^2 - 8v_{28} - 8v_{28}^2 - 4v_{30} - 8v_{30}^2 + 8v_{32} - 8v_{32}^2 - 2v_{35} - 6v_{37} \\
&- 12v_{54} - 8v_{54}^2 + v_{77} \\
S_{64} = &- v_4v_{104} - v_{79}v_{82}v_{104} + v_{80}v_{179} + v_{82}v_{104} - v_{83} \\
S_{65} = &- v_{17}v_{85}v_{104} + v_{84}v_{179} - v_{86} \\
S_{66} = &- v_{17}v_{88}v_{104} + v_{87}v_{179} - v_{89} \\
S_{67} = &- v_{17}v_{91}v_{104} + v_{90}v_{179} - v_{92} \\
S_{68} = &- v_{17}v_{94}v_{104} + v_{93}v_{179} - v_{95} \\
S_{69} = &- v_{17}v_{98}v_{104} + v_{96}v_{180} - v_{99} \\
S_{70} = &- v_{17}v_{101}v_{104} + v_{100}v_{180} - v_{102} \\
S_{71} = &- v_{17}v_{104}v_{105} + v_{103} - v_{106} \\
S_{72} = &- v_{17}v_{104}v_{108} + v_{107} - v_{109} \\
S_{73} = &- v_{17}v_{104}v_{111} + v_{110}v_{179} - v_{112}
\end{aligned}
$$

$$
\begin{aligned}
S_{74} =& -v_{17}v_{104}v_{114} + v_{113}v_{179} - v_{115} \\
S_{75} =& 4 + 2v_0 + 3v_0^2 - 6v_0^2v_{117} + 2v_0v_3 - 4v_0v_3v_{117} - 4v_0v_{117} \\
&+ v_1^2 - 2v_1^2v_{117} + 2v_3 + 2v_3^2 - 4v_3^2v_{117} - 4v_3v_{117} + v_4^2 \\
&- 2v_4^2v_{117} + 2v_5^2 - 4v_5^2v_{117} - 2v_8 + v_8^2 - 2v_8^2v_{117} \\
&+ 4v_8v_{117} - 2v_9v_{10} + 4v_9v_{10}v_{117} + v_{11}^2 - 2v_{11}^2v_{117} + v_{12}^2 \\
&- 2v_{12}^2v_{117} + v_{13}^2 - 2v_{13}^2v_{117} + v_{14}^2 - 2v_{14}^2v_{117} + v_{15}^2 \\
&- 2v_{15}^2v_{117} + v_{16}^2 - 2v_{16}^2v_{117} + v_{17}^2 - 2v_{17}^2v_{117} + v_{22} \\
&- 2v_{22}v_{117} + v_{77}^2 - 2v_{77}^2v_{117} - 2v_{79} + 2v_{79}^2 - 4v_{79}^2v_{117} \\
&+ 4v_{79}v_{117} + v_{80}^2 - 2v_{80}^2v_{117} + v_{81}^2 - 2v_{81}^2v_{117} + v_{84}^2 \\
&- 2v_{84}^2v_{117} + v_{87}^2 - 2v_{87}^2v_{117} + v_{90}^2 - 2v_{90}^2v_{117} + v_{93}^2 \\
&- 2v_{93}^2v_{117} + v_{96}^2 - 2v_{96}^2v_{117} + v_{97}^2 - 2v_{97}^2v_{117} + v_{100}^2 \\
&- 2v_{100}^2v_{117} + v_{103}^2 - 2v_{103}^2v_{117} + v_{104}^2 - 2v_{104}^2v_{117} + v_{107}^2 \\
&- 2v_{107}^2v_{117} + v_{110}^2 - 2v_{110}^2v_{117} + v_{113}^2 - 2v_{113}^2v_{117} - 8v_{117} \\
&- 2v_{117}v_{165}^2 + v_{118} + v_{165}^2 \\
S_{76} =& -3v_0^2v_{118} - 2v_0v_3v_{118} - 2v_0v_{118} - v_1^2v_{118} - 2v_3^2v_{118} \\
&- 2v_3v_{118} - v_4^2v_{118} - 2v_5^2v_{118} - v_8^2v_{118} + 2v_8v_{118} \\
&+ 2v_9v_{10}v_{118} - v_{11}^2v_{118} - v_{12}^2v_{118} - v_{13}^2v_{118} - v_{14}^2v_{118} \\
&- v_{15}^2v_{118} - v_{16}^2v_{118} - v_{17}^2v_{118} - v_{22}v_{118} - v_{77}^2v_{118} \\
&- 2v_{79}^2v_{118} + 2v_{79}v_{118} - v_{80}^2v_{118} - v_{81}^2v_{118} - v_{84}^2v_{118} \\
&- v_{87}^2v_{118} - v_{90}^2v_{118} - v_{93}^2v_{118} - v_{96}^2v_{118} - v_{97}^2v_{118} \\
&- v_{100}^2v_{118} - v_{103}^2v_{118} - v_{104}^2v_{118} - v_{107}^2v_{118} - v_{110}^2v_{118} \\
&- v_{113}^2v_{118} - 5v_{118} - v_{118}v_{165}^2 + v_{119} \\
S_{77} =& 2v_{116} + v_{116}^2 + v_{118}^2 - v_{119}^2 \\
S_{78} =& -v_{116}v_{121} - v_{121} + v_{122} \\
S_{79} =& -1 + v_{120}^2 + 2v_{120}v_{122} + v_{121}^2 \\
S_{80} =& -v_3^2v_{123} + 2v_3v_{116}v_{123} - v_3v_{121} + v_5 + 2v_{116}v_{123} \\
&- v_{120} - v_{121} \\
S_{81} =& -v_4^2v_{124} + 2v_4v_{116}v_{124} + 2v_4v_{124} - v_4v_{154} + v_{17} - v_{124} \\
&- v_{153} \\
S_{82} =& -1 + v_{79} + 2v_{116}v_{125}v_{165} - v_{125} + 2v_{125}v_{165} - v_{125}v_{165}^2 \\
&- v_{153} - v_{154}v_{165} \\
S_{83} =& -v_{116}v_{127} - v_{127} + v_{128} \\
S_{84} =& -1 + v_{126}^2 + 2v_{126}v_{128} + v_{127}^2 \\
S_{85} =& 1 + v_{116} - v_{116}v_{130} - v_{130} + v_{131} \\
S_{86} =& v_{129}^2 + 2v_{129}v_{131} - 2v_{130} + v_{130}^2 \\
S_{87} =& -v_{116}v_{133} - v_{133} + v_{134} \\
S_{88} =& -1 + v_{132}^2 + 2v_{132}v_{134} + v_{133}^2 \\
S_{89} =& -v_{116}v_{136} - v_{136} + v_{137} \\
S_{90} =& -1 + v_{135}^2 + 2v_{135}v_{137} + v_{136}^2 \\
S_{91} =& -v_{116}v_{139} - v_{139} + v_{140} \\
S_{92} =& -1 + v_{138}^2 + 2v_{138}v_{140} + v_{139}^2 \\
S_{93} =& -v_{116}v_{142} - v_{142} + v_{143} \\
S_{94} =& -1 + v_{141}^2 + 2v_{141}v_{143} + v_{142}^2
\end{aligned}
$$

$$
\begin{aligned}
S_{95} &= -v_{116}v_{145} - v_{145} + v_{146} \\
S_{96} &= -1 + v_{144}^2 + 2v_{144}v_{146} + v_{145}^2 \\
S_{97} &= -v_{116}v_{148} - v_{148} + v_{149} \\
S_{98} &= -1 + v_{147}^2 + 2v_{147}v_{149} + v_{148}^2 \\
S_{99} &= -v_{116}v_{151} - v_{151} + v_{152} \\
S_{100} &= -1 + v_{150}^2 + 2v_{150}v_{152} + v_{151}^2 \\
S_{101} &= 1 - v_{116}v_{154} - v_{154} + v_{155} \\
S_{102} &= -1 + 2v_{153} + v_{153}^2 + 2v_{153}v_{155} + v_{154}^2 \\
S_{103} &= -v_{116}v_{157} - v_{157} + v_{158} \\
S_{104} &= -1 + v_{156}^2 + 2v_{156}v_{158} + v_{157}^2 \\
S_{105} &= -v_{116}v_{160} - v_{160} + v_{161} \\
S_{106} &= -1 + v_{159}^2 + 2v_{159}v_{161} + v_{160}^2 \\
S_{107} &= -v_{116}v_{163} - v_{163} + v_{164} \\
S_{108} &= -1 + v_{162}^2 + 2v_{162}v_{164} + v_{163}^2 \\
S_{109} &= v_4 - 2v_4v_8 - 2v_4v_9 - 2v_4v_{10} - 2v_4v_{11} - 2v_4v_{12} \\
&\quad - 2v_4v_{13} - 2v_4v_{14} - 2v_4v_{15} - 2v_4v_{16} + v_{165} \\
S_{110} &= -16 + v_0 - v_{166}^2 \\
S_{111} &= -v_{116}v_{168} - v_{168} + v_{176} \\
S_{112} &= -1 + v_{167}^2 + v_{168}^2 - v_{176}^2 \\
S_{113} &= -v_{116}v_{170} - v_{167}v_{170}v_{173} - v_{170} + v_{177} \\
S_{114} &= -1 + v_{169}^2 + v_{170}^2 - v_{177}^2 \\
S_{115} &= -v_{163}^2 - 2v_{163}^2v_{172} + v_{168} \\
S_{116} &= v_{116} + v_{167}v_{173} - 4v_{168} - 8v_{168}v_{171} \\
S_{117} &= -v_{162} - v_{164} - v_{167}v_{174} + v_{169} \\
S_{118} &= -v_{77} - 4v_{163}v_{175} + v_{170} \\
S_{119} &= 26 + 2v_0^2 + 2v_0v_3 - 2v_0v_{121} - 2v_0v_{127} + v_1^2 - 2v_1v_{154} \\
&\quad + v_3^2 - 2v_3v_{127} + v_5^2 - 2v_5v_{145} + v_8^2 - 2v_8v_{130} + v_9^2 \\
&\quad - 2v_9v_{133} + v_{10}^2 - 2v_{10}v_{136} + v_{11}^2 - 2v_{11}v_{139} + v_{12}^2 \\
&\quad - 2v_{12}v_{142} + v_{13}^2 - 2v_{13}v_{148} + v_{14}^2 - 2v_{14}v_{151} + v_{15}^2 \\
&\quad - 2v_{15}v_{157} + v_{16}^2 - 2v_{16}v_{160} + v_{77}^2 - 2v_{77}v_{163} - v_{116}v_{178} \\
&\quad - 2v_{120} - 2v_{122} + v_{122}^2 - 2v_{126} - 2v_{128} + v_{128}^2 - 2v_{129} + 2v_{130} \\
&\quad - 2v_{131} + v_{131}^2 - 2v_{132} - 2v_{134} + v_{134}^2 - 2v_{135} - 2v_{137} + v_{137}^2 \\
&\quad - 2v_{138} - 2v_{140} + v_{140}^2 - 2v_{141} - 2v_{143} + v_{143}^2 - 2v_{144} - 2v_{146} \\
&\quad + v_{146}^2 - 2v_{147} - 2v_{149} + v_{149}^2 - 2v_{150} - 2v_{152} + v_{152}^2 - 2v_{153} \\
&\quad + v_{155}^2 - 2v_{156} - 2v_{158} + v_{158}^2 - 2v_{159} - 2v_{161} + v_{161}^2 - 2v_{162} \\
&\quad - 2v_{164} + v_{164}^2 \\
S_{120} &= -v_{81}v_{179} + v_{104} \\
S_{121} &= -v_{97}v_{180} + v_{104} \\
S_{122} &= -2 - v_{77}^2 - v_{121}^2 - v_{127}^2 + 2v_{130} - v_{130}^2 - v_{133}^2 - v_{136}^2 - v_{139}^2 \\
&\quad - v_{142}^2 - v_{145}^2 - v_{148}^2 - v_{151}^2 - v_{154}^2 - v_{157}^2 - v_{160}^2 + v_{163}^2 \\
&\quad - v_{181}^2 - v_{182}^2 - v_{183} - v_{183}^2 \\
S_{123} &= -1 - 4v_{83}^2 - 4v_{86}^2 - 4v_{89}^2 - 4v_{92}^2 - 4v_{95}^2 - 4v_{99}^2 - 4v_{102}^2 + v_{104}^2 - 4v_{106}^2 \\
&\quad - 4v_{109}^2 - 4v_{112}^2 - 4v_{115}^2 - v_{184}^2 - v_{185}^2 - v_{186} - v_{186}^2 - 4v_{188}^2
\end{aligned}
$$

$$
\begin{aligned}
S_{124} =\ & -v_4v_{187}+v_{104}\\
S_{125} =\ & v_2v_{104}-v_5v_{187}-v_{188}\\
S_{126} =\ & 1+v_{77}^2+v_{79}^2v_{189}-2v_{79}v_{80}v_{127}+2v_{79}v_{84}-2v_{79}v_{84}v_{130}\\
& -2v_{79}v_{87}v_{133}-2v_{79}v_{90}v_{136}-2v_{79}v_{93}v_{139}-2v_{79}v_{96}v_{142}\\
& -2v_{79}v_{100}v_{148}-2v_{79}v_{103}v_{151}-2v_{79}v_{107}v_{157}\\
& -2v_{79}v_{110}v_{160}-2v_{79}v_{113}v_{163}-2v_{79}v_{116}v_{189}\\
& +2v_{79}v_{126}v_{127}+2v_{79}v_{127}v_{128}-2v_{79}v_{129}+2v_{79}v_{129}v_{130}\\
& +2v_{79}v_{130}v_{131}-2v_{79}v_{131}+2v_{79}v_{132}v_{133}+2v_{79}v_{133}v_{134}\\
& +2v_{79}v_{135}v_{136}+2v_{79}v_{136}v_{137}+2v_{79}v_{138}v_{139}\\
& +2v_{79}v_{139}v_{140}+2v_{79}v_{141}v_{142}+2v_{79}v_{142}v_{143}\\
& +2v_{79}v_{147}v_{148}+2v_{79}v_{148}v_{149}+2v_{79}v_{150}v_{151}\\
& +2v_{79}v_{151}v_{152}+2v_{79}v_{156}v_{157}+2v_{79}v_{157}v_{158}\\
& +2v_{79}v_{159}v_{160}+2v_{79}v_{160}v_{161}+2v_{79}v_{162}v_{163}\\
& +2v_{79}v_{163}v_{164}-2v_{79}v_{189}+v_{80}^2-2v_{80}v_{126}+v_{84}^2-2v_{84}v_{129}\\
& +v_{87}^2-2v_{87}v_{132}+v_{90}^2-2v_{90}v_{135}+v_{93}^2-2v_{93}v_{138}+v_{96}^2\\
& -2v_{96}v_{141}+v_{100}^2-2v_{100}v_{147}+v_{103}^2-2v_{103}v_{150}+v_{107}^2\\
& -2v_{107}v_{156}+v_{110}^2-2v_{110}v_{159}+v_{113}^2-2v_{113}v_{162}+v_{121}^2\\
& +v_{126}^2+v_{129}^2+v_{132}^2+v_{135}^2+v_{138}^2+v_{141}^2+v_{145}^2+v_{147}^2\\
& +v_{150}^2+v_{154}^2+v_{156}^2+v_{159}^2+v_{162}^2-2v_{163}^2+v_{181}^2+v_{182}^2\\
& +v_{183}+v_{183}^2+v_{189}\\
S_{127} =\ & v_{79}^2v_{190}-2v_{79}v_{81}v_{121}-2v_{79}v_{97}v_{145}-2v_{79}v_{104}v_{154}\\
& -2v_{79}v_{116}v_{190}+2v_{79}v_{120}v_{121}+2v_{79}v_{121}v_{122}\\
& +2v_{79}v_{144}v_{145}+2v_{79}v_{145}v_{146}+2v_{79}v_{153}v_{154}+2v_{79}v_{154}\\
& +2v_{79}v_{154}v_{155}-4v_{79}v_{190}+v_{81}^2-2v_{81}v_{120}+2v_{81}v_{121}+v_{97}^2\\
& -2v_{97}v_{144}+2v_{97}v_{145}+v_{104}^2-2v_{104}v_{153}+2v_{104}v_{154}\\
& +2v_{116}v_{190}+v_{120}^2-2v_{120}v_{121}-v_{121}^2-2v_{121}v_{122}+v_{144}^2\\
& -2v_{144}v_{145}-v_{145}^2-2v_{145}v_{146}+v_{153}^2-2v_{153}v_{154}-2v_{154}\\
& -v_{154}^2-2v_{154}v_{155}+4v_{190}
\end{aligned}
$$

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